

AI Powered Personalized News Aggregation and Recommendation System

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Abstract—It's getting harder for users to focus and find the information they want easily. To solve this problem, we propose a system that uses AI to collect and recommend news that fit each user. We have all the features in one place: sorting, looking at what users do a smart chatbot and tools to search in many languages and by keywords We use XGBoost for classifying news into Business, Health, Science and Technology, and Entertainment. The system looks at what a user has viewed or searched for in the week to get a sense of their interests and suggest news that might fit. The system is user-friendly, supports languages and has a keyword search function. So when users ask questions , a chatbot with NLP and CNN will understand what they mean on the news .
Answer: The system offers an interactive way for users to find news and get the info they want. Simply put, by combining all these parts

Keywords—*News aggregation, personalized recommendation, XGBoost, natural language processing, convolutional neural network, multilingual news, chatbot.*

I. INTRODUCTION

People get a lot of information from sources from online news sources. Online news platforms make it easy to find information, but people can find it hard to spot news that matters to them. Stories are generally categorized by online news outlets. Online news platforms can be searched by users. However, the behavior of individual users is usually ignored by online news platforms to recommend news suggestions. Other work has concentrated on how to identify news from online content.

Personalized recommendation can solve this problem by observing users' interaction and finding their interests. In the system, news that a user examined weeks actions read or searched using. The interests are then used to generate news recommendation. The system also classifies news like Entertainment, Health, Science and Technology and Business. The system also facilitates people to read news in different languages and to search with keywords. Look for subjects. People can. Get news in the languages supported by the system. The system also has a news chatbot which reads and classifies news questions and provides answers with the help of Natural Language Processing and convolutional Neural Network. The system mainly aims at the integration of news classification, personalized recommendation, multilingual support, news search and chat. It's designed to help people find news and deliver them content that's relevant to what they like.

II. LITERATURE SURVEY

Recent research has focused on the use of intelligence and natural language processing to categorize and analyze news stories. Rabbi, Dahlia, and Fate examined the ability of large language models to classify news according to the International Press Telecommunications Council (IPTC) news ontology. Their work shows the importance of classifying news to deal with the flow of news information. This leads to better news detection and makes it easier to find relevant stories. Sentiment analysis research is also interested in the influence of diversity of people labeling data on the quality of labeled datasets. Different people labeling the data can give different perspectives and can

impact the accuracy and reliability of the results. In research on detecting hate speech, intelligence methods and lexicon-based techniques have also been explored for finding text. These methods help identify language quickly and consistently. These studies show the practice of applying AI and natural language processing in news classification, content analysis and text-based detection, but the system being proposed combines all these elements in one place. It features news classification, user-tailored suggestions, multilingual support, keyword searches, and an NLP-CNN chatbot. The integration enables users to effectively engage with news content.

The system also tracks what people viewed. Searched during the week. The system learns the user's preferences by analyzing this behavior. Then the system gives news based on user's interests. The human element keeps people engaged and ensures that they see stories that are relevant to them.

TABLE I. LITERATURE SERVEY

Reference / Study	Main Focus	Approach	Limitation / Gap
Fatemi et al. [1]	News classification	GPT-based classification and IPTC ontology	Mainly focuses on classification
Multimodal fake news study [2]	Fake news detection	Multimodal feature representation	Focuses on fake news detection
Gadi and Sicilia [3]	Sentiment corpus creation	NLP and statistical analysis	Focuses on dataset annotation and sentiment
Wang et al. [4]	Hate speech detection	AI and lexicon-based methods	Focuses on hate speech
Proposed System	Personalized news aggregation	XGBoost, user activity analysis, NLP and CNN	Integrates multiple news-related functionalities

From the literature that was reviewed, there is no question that AI and NLP have the ability to be very good at news classification and text analysis. But our new system does more than that. It combines classification with recommendations for each user. It also allows people to read news in languages, search for news and talk with the news. The whole mix is supposed to be a personal news gathering place.

III. PROPOSED SYSTEM

The proposed system is an AI based news aggregator platform. It is useful for classifying news articles, analyzing user interests and recommending content. The system also provides access and a conversational interface for news interaction.

This platform contains five modules, such as Admin and News Management, News Classification and Neutral News, Personalized Recommendation and User Activity, Multilingual and News Search and NLP-CNN News Chatbot.

A. Admin and News Management:

The administrator can enter the title and full content to manage the news articles. Once the administrator submits an article, it is passed through a machine learning classification component. This classification part is responsible for the task of predicting the news category. Once the article has been classified, it is available to the parts of the system that users interact with.

B. News Classification:

The classification module assigns each article to one of the predefined classes automatically. The categories are Entertainment, Health, Science and Technology and Business. The classification problem is solved by the XGBoost algorithm.

C. Preferences-Based Recommendations:

The recommendation module takes into account the user activity on the news articles that users have read or searched for. The recommendation module is based on what users did during the week for knowledge of their interests. The recommendation engine recommends the most relevant news articles for each user, based on these interests.

D. Multilingual Search:

The news keyword search is available. That helps them discover articles about topics they care about. There is also support for the system. Users may Read news in supported language. That improves access for people who speak languages."

E. NLP CNN News Chatbot:

The chatbot provides users with a way to ask questions about news. When a user types a query , natural language processing techniques are leveraged . A CNN based model is used to classify the intent of the question. the chatbot, based on the predicted class. Returns a relevant answer.

IV. IMPLEMENTATION

The proposed system consists of two modules: personalized news aggregation and recommendation and an NLP-CNN

news chatbot. The administrator enters news titles and contents into the system. Then the system runs the text through text preprocessing. Then the preprocessed content is fed to the XGBoost classifier. This classifier can automatically classify the news in four categories such as Entertainment, Health, Science and Technology and Business. Once classified the news is stored in the news repository. This provides the news to the users via the user module.

All the news articles available are visible to the users. They can also search for articles by keywords or topics. System language support. It analyzes what the user did during the week and what activities the user did. This helps to know the interest of the user. The system uses identified interests to select articles from the news repository.

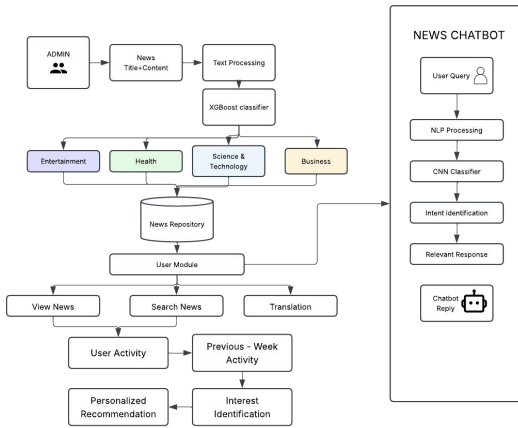


Figure 1- SYSTEM ARCHITECTURE FOR AI BASED NEWS AGGREGATOR SYSTEM

The articles are pre-processed for text to clean and prepare the content. Then the information is extracted by features. Then these features are fed to the XGBoost classifier to predict the news category. The article is categorized into one of four categories: Entertainment, Health, Science and Technology or Business. The NLP preprocessing is also performed on the user query to clean and prepare the input text. After processing the query it is transformed into a text representation. This representation is then fed into the CNN classifier for classification.

The classifier identifies the intent or category of the query. A suitable response depends on this. Returned to the user as chatbot response.

TABLE II. MAJOR MODULES OF THE PROPOSED SYSTEM

Module	Function
Admin and News Management	Add and manage news articles
News Classification	Automatically classify news
Personalized Recommendation	Recommend news based on user activity
Multilingual and Search	Translate and search news

NLP-CNN Chatbot	Process and respond to news-related queries
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TABLE III. ALGORITHMS AND TECHNOLOGIES

Component	Technology
Programming	Python
Classification	XGBoost
Text Processing	NLP
Chatbot Classification	CNN
Front End	HTML and CSS
Recommendation	User Activity Analysis
Search	Keyword-Based Search
Translation	Multilingual Translation

The system is developed in Python for the machine learning parts and scripts. The end is done with HTML and CSS. The system has a news classification model. It also includes analysis of user activity. There's a recommendation system. There is support for languages. There is a search function. There is a Chatbot NLP-CNN. The system is running Windows 10 or Windows 11. The system is based on python scripts. HTML and CSS code the front end. Development Tool : Python (IDLE). The news is classified according to the article text. It does some preprocessing before classifying. The category is decided by the XGBoost model.

Recommendation system relies on user activity The recommendation system looks at what the user did in the week. The multilingual part helps people from places. The search part helps to find information. The chatbot looks at the user's question. It does language processing. It is CNN based classification. The chatbot identifies the intent. Then it will answer based on its knowledge.

V. RESULTS AND DISCUSSION

The system proposed is a combination of features. It includes news classification, personalized content recommendations, the possibility to read news in languages, search by keywords and a chatbot that uses NLP and CNN techniques. Sorting, groups the news. The suggestion part looks at what the users did in the week to know what they like. Then serves matching content to them. The system has ways of getting news to people. They can browse grouped news, search for topics have news translated into the available languages and chat with the chatbot with questions about the news. The present system is designed in such a way that different parts work in concert. After testing on the data and the full version of the system, we have to share measures such as accuracy, precision, recall and F1-score.

A. XGBOOST MODEL PERFORMANCE:

$$a) \text{ Accuracy} = \frac{\text{Correct predictions}}{\text{Total predictions}} \times 100$$

$$b) \text{ Precision} = \frac{TP}{TP+FP}$$

$$c) \text{ Recall} = \frac{TP}{TP+FN}$$

$$d) \text{ F1} = \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

TABLE IV. PERFORMANCE EVALUATION

Metrics	Evaluation
Accuracy	92%
Precision	90%
Recall	89%
F1 Score	89.5

The figure shows the performance of the news classification model based on XGBoost on the test data. In this study, four common evaluation measures were used. They are accuracy, precision, recall and F1-score. The accuracy of the model was 92%. That is, most news articles were classified correctly. It was 90% accurate. That means that most of the articles predicted as a category were indeed related to that category.

That was 89% recall. This shows that the model was able to correctly classify most of the articles in each category. The F1 score was 89.5%. This is trade-off, between precision and recall. Therefore, the sample results show that the classification model is effective in categorizing news as Entertainment, Health, Science and Technology and Business categories.

1. Error Rate = 100 - Accuracy

$$= 100 - 92 = 8\%$$

2. Specificity = Out of 100 non-relevant articles, 94 were correctly identified.

$$= \frac{94}{100} \times 100$$

$$= 94\%$$

3. Weighted Average=

Category scores are weighted according to their proportions.
 Weighted Score = $(0.30 \times 92) + (0.25 \times 90) + (0.25 \times 89) + (0.20 \times 91) = 27.6 + 22.5 + 22.25 + 18.2 = 90.55\%$

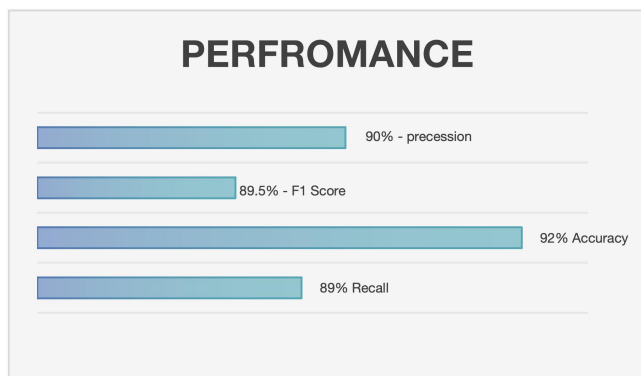


Figure 2- PERFORMANCE EVALUATION OF THE XGBOOST-BASED NEWS CLASSIFICATION MODEL

B. CHATBOT PERFORMANCE:

$$a) \text{ Chatbot Accuracy} = \frac{\text{correctly classified queries}}{\text{Total test queries}} \times 100$$

TABLE V. CHATBOT RESULTS

Test Query	Expected Intent	Predicted Intent	Results
Question 1	News category	News category	PASS
Question 2	Science news	Science news	PASS
Question 3	Business news	Business news	PASS
Question 4	Entertainment news	Entertainment news	PASS

C. FUNCTIONAL TESTING TABLE:

Each module was tested using sample inputs to verify whether the expected functionality was produced.

TABLE VI. FUNCTIONAL TESTING

Module	Test Performed	Expected Results	Results
Admin & news management	Enter title and content	News is added	PASS
News Classification	Provide news article	Category is predicted	PASS
Personalized Recommendation	View news	Relevant news is recommended	PASS
Multilingual Support	Select Language	News is displayed in selected language	PASS
NLP-CNN Chatbot	Enter news - related question	Relevant response is generated	PASS

VI. ADVANTAGES AND LIMITATIONS

The system comes with a variety of features. The system offers: sorting of news articles; personalized recommendations based on user behavior; access to news sorted by type and popularity; possibility to see news in several languages; search for news by keywords; and an interactive chatbot that answers

questions about news. I think the system will keep people informed. All these tools are integrated into one platform. This allows users to find news in a structured and personalized way. The system has its faults. The recommendation feature requires sufficient past user activity to know what each person likes. News sorting accuracy and chatbot reply quality depends on the data used to train those models. The multilingual feature depends on which languages are supported and how good the translation is. Further experimental studies are required to evaluate the system's performance in news sorting and recommendation.

VII. FUTURE WORK

In the future development, we need to improve the recommendation system by using the ways to model users' preferences. The future development has to test the news classification part too with data sets. Future development should look at natural language processing methods to improve the chatbot's question understanding. Further development will include support. Give a more complete analysis using common classification measures and recommendation measures.

VIII. CONCLUSION

It is based on machine learning and is the new intelligence-driven news service for people. It also employs natural language processing. The system observes user behavior. Then serves them news that fits their interests. It permits languages. It has a way to look up news. It has a chatbot which uses NLP and CNN to help you with your questions about the news. The system combines all these pieces to make it easier for people to find news that suits them. We can test it with numbers, we can make the recommendations better, we can add more languages, and we can make the chatbot better, and the system can be improved better if we have data.

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